KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · MATH

The Tide Pool Survey — Teacher Lesson Plan & Answer Key

Learning objectives: Students read line plots marked in halves, fourths and eighths, make a line plot from a list of measurements by choosing the scale and renaming equivalent fractions, add and subtract fractions with like and unlike denominators to answer questions about the data, find the fair share (the total shared equally among all the data points), and solve multistep survey problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Vikram (11), who joins a low-tide survey at Haystack Rock in Cannon Beach, Oregon, with his cousin Uzoma. Ms. Uchida, a volunteer, measures animals with a ruler marked in eighths without picking them up, and Mr. Varga, a retired science teacher, turns the lists into line plots. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs and a final period.
2. Slide 1 — Warm-Up, Part 1: Measurement Conversions Review (4 pts, 2 each, Week 22 skill). If a student misses both warm-up slides, review "bigger to smaller, multiply; smaller to bigger, divide."
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): mixed measurements, smaller-to-bigger units and a multistep conversion.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W23.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why a Tide Pool Survey Runs on Data: nonfiction connection about Haystack Rock, a basalt sea stack 235 feet tall. Its tide pools are a protected Marine Garden where nothing may be taken, the rock is part of the Oregon Islands National Wildlife Refuge, and tufted puffins nest there in spring and summer. Introduces the intertidal zone. No questions.
6. Slide 5 — Picture It: Reading a Line Plot. Twelve purple shore crab widths in eighths of an inch: count every X (12), find the tallest stack (1 3/8 inches), count the crabs wider than 1 1/2 inches (3; the ones at exactly 1 1/2 do not count), and subtract widest minus narrowest (1 7/8 − 1 1/8 = 3/4 inch).
7. Slide 6 — Your Turn, Part 1: Reading Line Plots (8 pts, 4 each): a different plot in each set (limpets, black turban snails, aggregating anemones). Stress "longer than" versus "or wider."
8. Slide 7 — Your Turn, Part 2 (4 pts): the total of two stacks (multiply each value by its number of X's, then add).
9. Slide 8 — Teach: Making a Line Plot. Vikram's 10 hermit crab shells: find the least and greatest, choose eighths (the smallest unit in the data), rename equivalent fractions (6/8 = 3/4, 1 2/8 = 1 1/4) so they share a stack, and count the X's to check against the list.
10. Slide 9 — Your Turn, Part 1: Making Line Plots (8 pts, 4 each): how many X's go in a stack after renaming, and which scale to use.
11. Slide 10 — Your Turn, Part 2 (4 pts): plot a list of 10 measurements on scrap paper, then answer.

12. Slide 11 — Teach: Solving Problems with Line Plot Data. Common denominator first: four 1 3/8-inch crabs = 5 1/2 inches; 1 7/8 − 1 2/8 = 5/8 inch; rename 2 as 1 8/8, so 1 8/8 − 1 3/8 = 5/8 inch. “All the crabs at 1 1/2 inches” means every X in that stack.
13. Slide 12 — Your Turn, Part 1: Adding and Subtracting the Data (8 pts, 4 each): a combined length and a “how much longer” difference.
14. Slide 13 — Your Turn, Part 2 (4 pts): compare the totals of two stacks, regrouping a whole when needed.
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (line plot, data, tick mark, scale, equivalent fractions, common denominator, fair share, intertidal zone).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Finding the Fair Share. Ms. Uchida’s 8 jars hold 32/8 = 4 cups in all, so the fair share is 4 ÷ 8 = 1/2 cup. Divide by the number of X’s (8), not the number of stacks (5). Check: the 5/8 cup of extra above 1/2 cup exactly fills the 5/8 cup of gaps below it.
18. Slide 17 — Your Turn: Fair Shares (6 pts, 3 each): a fair share from a plot, and working backward from a fair share to the total.
19. Slide 18 — Real World: Volunteer Shifts (4 pts): a different plot of shift lengths in each set: combine two stacks, compare stack totals, or find the fair share for 8 volunteers.
20. Slide 19 — Real World: Collecting and Cleaning Up (4 pts): how much more seawater is needed to reach 4 cups, how many more pounds of litter reach an 8-pound goal, or the total of the walks shorter than 1/2 mile.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a “which explanation is correct?” item and a “find the mistake” item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, and one grammar fix). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences, same rules. Both serial-comma and no-serial-comma forms of a list are accepted where a list appears.
24. Slide 23 — Match-Up Round (10 pts): 5 fraction problems (sums, differences, products and fair shares) matched to their answers, 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick items against the student’s own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: The Survey Report (8 pts, 4 each): a different line plot in each set (10 crab widths, 8 seawater jars, or 9 sea star count walks): compare stack totals, find a total or fair share, or find the room left in a 4-cup bottle.
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W23.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Survey Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Cannon Beach Data Champion!” with the congratulations message “You finished the whole tide pool survey at Haystack Rock!” and a big confetti burst; below 85 shows “Good Work at the Tide Pools!” with encouragement (“You are getting closer every time.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: MEASUREMENT CONVERSIONS REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Fractions of a unit (Week 22): a tow line on the tugboat is 2 2/3 yards long. How many feet long is it? Feet:	**8** (also accepted: 8 feet, 8 foot, 8 ft)	Each yard is 3 feet: 2 × 3 = 6, and 2/3 of 3 = 2. 6 + 2 = 8 feet.
r2_s0	Set 1: Comparing (Week 22): which is heavier, a 2 3/4-pound bag of bolts or a 46-ounce bag of bolts? [46 ounces, because 2 3/4 pounds = 44 ounces / 2 3/4 pounds, because pounds are bigger units / They weigh the same / 46 ounces, because 2 3/4 pounds = 40 ounces]	**46 ounces, because 2 3/4 pounds = 44 ounces**	1 pound = 16 ounces. 2 × 16 = 32, and 3/4 of 16 = 12. 32 + 12 = 44 ounces, and 46 ounces is 2 ounces more.
r3_s0	Set 1: Mixed measurements (Week 22): a fender rope is 3 feet 8 inches long. How many inches is that? Inches:	**44** (also accepted: 44 inches, 44 inch, 44 in)	3 feet = 3 × 12 = 36 inches. Add the extra 8 inches: 36 + 8 = 44 inches.
r4_s0	Set 1: Smaller to bigger (Week 22): a box of nails has a mass of 1,750 grams. How many kilograms is that? Kilograms:	**1.75** (also accepted: 1.75 kilograms, 1.75 kilogram, 1.75 kg, 1 3/4, 1 3/4 kilograms, 1 3/4 kg)	Smaller to bigger, divide: 1,750 ÷ 1,000 = 1.75 kilograms. The decimal point moves 3 places to the left.
r1_s1	Set 2: Metric (Week 22): a ship travels 4.5 kilometers across the harbor. How many meters is that? Meters:	**4,500** (also accepted: 4,500 meters, 4,500 meter, 4,500 m)	Bigger to smaller, multiply: 4.5 × 1,000 = 4,500 meters. The decimal point moves 3 places to the right.
r2_s1	Set 2: Comparing (Week 22): which is longer, 1 1/2 yards of rope or 50 inches of rope? [1 1/2 yards, because it equals 54 inches / 50 inches, because 50 is greater than 1 1/2 / They are the same length / 50 inches, because 1 1/2 yards = 45 inches]	**1 1/2 yards, because it equals 54 inches**	1 yard = 36 inches. 36 + 18 = 54 inches, which is 4 inches more than 50 inches.
r3_s1	Set 2: Smaller to bigger (Week 22): a sack of rice weighs 54 ounces. How many pounds is that? (You may use a mixed number.) Pounds:	**3 3/8** (also accepted, with or without the unit word: 3 6/16, 27/8, 3-3/8, 3 and 3/8, 3.375, 3 pounds 6 ounces, 3 lb 6 oz, 3 pounds and 6 ounces)	54 ÷ 16 = 3 R6, so 54 ounces = 3 pounds 6 ounces. 6/16 = 3/8, so that is 3 3/8 pounds.
r4_s1	Set 2: Three steps (Week 22): Uncle Torvald has a 20-yard line. He cuts 3 pieces that are each 12 feet long. How many feet of line are left? Feet:	**24** (also accepted: 24 feet, 24 foot, 24 ft)	The line is 20 × 3 = 60 feet. The pieces use 3 × 12 = 36 feet. 60 − 36 = 24 feet left.
r1_s2	Set 3: Capacity (Week 22): the galley makes 3 1/2 gallons of soup. How many quarts is that? Quarts:	**14** (also accepted: 14 quarts, 14 quart, 14 qt)	Each gallon is 4 quarts: 3 × 4 = 12, and 1/2 of 4 = 2. 12 + 2 = 14 quarts.
r2_s2	Set 3: Comparing (Week 22): which bottle holds more, 2.5 liters or 2,400 milliliters? [2.5 liters, because it equals 2,500 milliliters / 2,400 milliliters, because 2,400 is a bigger number / They hold the same amount / 2,400 milliliters, because 2.5 liters = 250 milliliters]	**2.5 liters, because it equals 2,500 milliliters**	1 liter = 1,000 milliliters, so 2.5 liters = 2,500 milliliters, which is 100 milliliters more than 2,400.
r3_s2	Set 3: Metric (Week 22): a crate lid is 0.8 meter wide. How many centimeters is that? Centimeters:	**80** (also accepted: 80 centimeters, 80 centimeter, 80 cm)	Bigger to smaller, multiply: 0.8 × 100 = 80 centimeters.

ID	Question	Answer	Why (explain to student)
r4_s2	Set 3: Two steps (Week 22): a crate has a mass of 2.3 kilograms, and its packing adds 650 grams. What is the total mass in grams? Grams:	**2,950** (also accepted: 2,950 grams, 2,950 gram, 2,950 g)	2.3 kilograms = 2.3 × 1,000 = 2,300 grams. 2,300 + 650 = 2,950 grams.

YOUR TURN: READING LINE PLOTS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: *Mr. Varga plotted the lengths of the limpets in one tide pool. [Line plot "Limpet Shell Lengths", Length (inches); X's: 5/8 (2); 3/4 (3); 7/8 (1); 1 (4); 1 1/8 (2); 1 3/8 (1).]* How many limpets are longer than 7/8 inch? Limpets:	**7** (also accepted: 7 limpets, 7 limpet)	Count the X's to the right of 7/8: 4 at 1 inch, 2 at 1 1/8, and 1 at 1 3/8. 4 + 2 + 1 = 7 limpets. The limpet at exactly 7/8 does not count.
l2_s0	Set 1: What is the difference between the longest and the shortest limpet? [3/4 inch / 2 inches / 1 inch / 5/8 inch]	**3/4 inch**	Longest: 1 3/8 = 11/8. Shortest: 5/8. 11/8 − 5/8 = 6/8 = 3/4 inch. (2 inches comes from adding instead of subtracting.)
l3_s0	Set 1: *[Line plot "Limpet Shell Lengths", Length (inches); X's: 5/8 (2); 3/4 (3); 7/8 (1); 1 (4); 1 1/8 (2); 1 3/8 (1).]* Two steps: what is the total length of all the limpets that measured 3/4 inch, together with all the limpets that measured 1 1/8 inches? Inches:	**4 1/2** (also accepted, with or without the unit word: 4 2/4, 4 4/8, 4 8/16, 9/2, 4-1/2, 4 and 1/2, 4.5)	Three limpets at 3/4: 3 × 3/4 = 9/4 = 2 1/4. Two limpets at 1 1/8: 2 × 1 1/8 = 2 2/8 = 2 1/4. Total: 2 1/4 + 2 1/4 = 4 1/2 inches.
l1_s1	Set 2: *Mr. Varga plotted the heights of the black turban snails in one tide pool. [Line plot "Black Turban Snail Shell Heights", Height (inches); X's: 7/8 (1); 1 (3); 1 1/8 (2); 1 1/4 (4); 1 3/8 (2); 1 3/4 (1).]* How many snails are shorter than 1 1/4 inches? Snails:	**6** (also accepted: 6 snails, 6 snail)	Count the X's to the left of 1 1/4: 1 at 7/8, 3 at 1, and 2 at 1 1/8. 1 + 3 + 2 = 6 snails. The snails at exactly 1 1/4 do not count.
l2_s1	Set 2: What is the difference between the tallest and the shortest snail? [7/8 inch / 2 5/8 inches / 1 inch / 3/4 inch]	**7/8 inch**	Tallest: 1 3/4 = 14/8. Shortest: 7/8. 14/8 − 7/8 = 7/8 inch. (2 5/8 comes from adding instead of subtracting.)
l3_s1	Set 2: *[Line plot "Black Turban Snail Shell Heights", Height (inches); X's: 7/8 (1); 1 (3); 1 1/8 (2); 1 1/4 (4); 1 3/8 (2); 1 3/4 (1).]* Two steps: what is the total height of all the snails that measured 1 inch, together with all the snails that measured 1 1/8 inches? Inches:	**5 1/4** (also accepted, with or without the unit word: 5 2/8, 5 4/16, 21/4, 5-1/4, 5 and 1/4, 5.25)	Three snails at 1 inch: 3 × 1 = 3. Two snails at 1 1/8: 2 × 1 1/8 = 2 1/4. Total: 3 + 2 1/4 = 5 1/4 inches.
l1_s2	Set 3: *Mr. Varga plotted the widths of the aggregating anemones in one tide pool, measured while they were closed. [Line plot "Aggregating Anemone Widths", Width (inches); X's: 1/4 (1); 1/2 (3); 5/8 (3); 3/4 (2); 1 (2); 1 1/4 (1).]* How many anemones are 5/8 inch wide or wider? Anemones:	**8** (also accepted: 8 anemones, 8 anemone)	Count the X's at 5/8 and to its right: 3 at 5/8, 2 at 3/4, 2 at 1, and 1 at 1 1/4. 3 + 2 + 2 + 1 = 8 anemones. "Or wider" means the ones at exactly 5/8 count.

ID	Question	Answer	Why (explain to student)
l2_s2	Set 3: What is the difference between the widest and the narrowest anemone? [1 inch / 1 1/2 inches / 3/4 inch / 7/8 inch]	**1 inch**	Widest: 1 1/4. Narrowest: 1/4. 1 1/4 − 1/4 = 1 inch. (1 1/2 comes from adding instead of subtracting.)
l3_s2	Set 3: *[Line plot "Aggregating Anemone Widths", Width (inches); X's: 1/4 (1); 1/2 (3); 5/8 (3); 3/4 (2); 1 (2); 1 1/4 (1).]* Two steps: what is the total width of all the anemones that measured 1/2 inch, together with all the anemones that measured 3/4 inch? Inches:	**3** (also accepted: 3 inches, 3 inch, 3 in)	Three anemones at 1/2: 3 × 1/2 = 1 1/2. Two anemones at 3/4: 2 × 3/4 = 1 1/2. Total: 1 1/2 + 1 1/2 = 3 inches.

YOUR TURN: MAKING LINE PLOTS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Uzoma's list of limpet lengths, in inches: 1/4, 5/8, 1/2, 3/4, 4/8, 5/8, 2/4, 7/8. When he makes a line plot, how many X's go above 1/2? X's above 1/2:	**3** (also accepted with a unit word, such as 3 x's or 3 marks, or 2 other unit spellings)	1/2, 4/8, and 2/4 are equivalent fractions, so all three go in the same stack above 1/2. That makes 3 X's.
k2_s0	Set 1: Uzoma's measurements are 1/2, 1 1/4, 3/4, 1 3/8, and 2 inches. Which tick marks does his number line need so every measurement sits on a tick? [Eighths (a tick every 1/8 inch) / Halves (a tick every 1/2 inch) / Fourths (a tick every 1/4 inch) / Wholes (a tick every 1 inch)]	**Eighths (a tick every 1/8 inch)**	1 3/8 is in eighths, so the number line needs eighths. Eighths also land on 1/2, 3/4, 1 1/4, and 2. Fourths would leave 1 3/8 between two ticks.
k3_s0	Set 1: Uzoma measured the widths of 10 closed sea anemones, in inches:1 1/2, 1 3/4, 1 2/4, 1 6/8, 2, 1 1/2, 1 7/8, 1 3/4, 1 5/8, 1 2/4. After he plots them, how many X's are in the tallest stack? X's:	**4** (also accepted with a unit word, such as 4 x's or 4 marks, or 2 other unit spellings)	Rename first: 1 2/4 = 1 1/2 and 1 6/8 = 1 3/4. The 1 1/2 stack has 1 1/2, 1 2/4, 1 1/2, 1 2/4 = 4 X's. The 1 3/4 stack has 1 3/4, 1 6/8, 1 3/4 = 3 X's. The tallest stack has 4.
k1_s1	Set 2: Vikram's list of snail heights, in inches: 3/8, 6/8, 1/2, 3/4, 1, 7/8, 3/4, 2/8, 6/8. When he makes a line plot, how many X's go above 3/4? X's above 3/4:	**4** (also accepted with a unit word, such as 4 x's or 4 marks, or 2 other unit spellings)	6/8 and 3/4 are equivalent fractions. The list has 6/8 two times and 3/4 two times, so 4 X's go in the stack above 3/4.
k2_s1	Set 2: Ms. Uchida's measurements are 1 1/2, 2, 2 1/4, 1 3/4, and 2 1/2 inches. Which scale puts every measurement on a tick mark AND uses the fewest tick marks? [Fourths (a tick every 1/4 inch) / Halves (a tick every 1/2 inch) / Eighths (a tick every 1/8 inch) / Wholes (a tick every 1 inch)]	**Fourths (a tick every 1/4 inch)**	2 1/4 and 1 3/4 are in fourths, so halves will not work. Eighths would also work, but fourths do the job with fewer tick marks.
k3_s1	Set 2: Ms. Uchida measured 10 chitons, in inches: 2, 1 7/8, 2 1/4, 2 2/8, 2 1/8, 2 1/4, 2 4/8, 2 1/2, 2 2/4, 1 7/8. After she plots them, how many X's sit to the right of 2? X's:	**7** (also accepted with a unit word, such as 7 x's or 7 marks, or 2 other unit spellings)	Greater than 2: 2 1/4, 2 2/8, 2 1/8, 2 1/4, 2 4/8, 2 1/2, 2 2/4. That is 7 X's. The X at exactly 2 and the two at 1 7/8 are not to the right of 2.
k1_s2	Set 3: Uzoma's list of crab widths, in inches: 1 1/4, 1 2/8, 1 1/2, 1 3/8, 1 1/4, 1 4/8, 1 5/8, 1 2/8. When he makes a line plot, how many X's go above 1 1/4? X's above 1 1/4:	**4** (also accepted with a unit word, such as 4 x's or 4 marks, or 2 other unit spellings)	1 2/8 and 1 1/4 are equivalent. The list has 1 1/4 two times and 1 2/8 two times, so 4 X's go above 1 1/4. (1 4/8 goes above 1 1/2 instead.)

ID	Question	Answer	Why (explain to student)
k2_s2	Set 3: Vikram's number line is marked in eighths, from 0 to 1. At which tick mark does the X for 3/4 go? [The tick for 6/8 / The tick for 3/8 / The tick for 4/8 / The tick for 7/8]	**The tick for 6/8**	3/4 = 6/8 (multiply the top and bottom by 2), so the X goes on the sixth tick mark after 0.
k3_s2	Set 3: Vikram measured the arms of 10 ochre sea stars to the nearest 1/4 inch: 3, 3 1/2, 3 2/4, 4, 3 3/4, 3 1/4, 3 2/4, 4, 3 6/8, 4 1/4. After he plots them, which length has the tallest stack of X's? Inches:	**3 1/2** (also accepted, with or without the unit word: 3 2/4, 3 4/8, 3 8/16, 7/2, 3-1/2, 3 and 1/2, 3.5)	Rename first: 3 2/4 = 3 1/2 and 3 6/8 = 3 3/4. The 3 1/2 stack has 3 1/2, 3 2/4, 3 2/4 = 3 X's. The 4 and 3 3/4 stacks have only 2 each. The tallest stack is at 3 1/2 inches.

YOUR TURN: ADDING AND SUBTRACTING THE DATA (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: *Use Mr. Varga's limpet line plot again. [Line plot "Limpet Shell Lengths", Length (inches); X's: 5/8 (2); 3/4 (3); 7/8 (1); 1 (4); 1 1/8 (2); 1 3/8 (1).]* What is the combined length of the two 5/8-inch limpets and the one 7/8-inch limpet? Inches:	**2 1/8** (also accepted, with or without the unit word: 2 2/16, 17/8, 2-1/8, 2 and 1/8, 2.125)	5/8 + 5/8 + 7/8 = 17/8. 17/8 = 2 1/8 inches.
e2_s0	Set 1: How much longer is a 1 1/8-inch limpet than a 3/4-inch limpet? [3/8 inch / 1/4 inch / 5/8 inch / 1 7/8 inches]	**3/8 inch**	Rename 3/4 as 6/8 and 1 1/8 as 9/8. 9/8 − 6/8 = 3/8 inch. (1 7/8 comes from adding.)
e3_s0	Set 1: Two steps: how much more is the total length of the four 1-inch limpets than the total length of the three 3/4-inch limpets? Inches:	**1 3/4** (also accepted, with or without the unit word: 1 6/8, 1 12/16, 7/4, 1-3/4, 1 and 3/4, 1.75)	Four 1-inch limpets: 4 inches. Three 3/4-inch limpets: 3 × 3/4 = 9/4 = 2 1/4 inches. Rename 4 as 3 4/4: 3 4/4 − 2 1/4 = 1 3/4 inches.
e1_s1	Set 2: *Use Mr. Varga's snail line plot again. [Line plot "Black Turban Snail Shell Heights", Height (inches); X's: 7/8 (1); 1 (3); 1 1/8 (2); 1 1/4 (4); 1 3/8 (2); 1 3/4 (1).]* What is the combined height of the two 1 3/8-inch snails and the one 7/8-inch snail? Inches:	**3 5/8** (also accepted, with or without the unit word: 3 10/16, 29/8, 3-5/8, 3 and 5/8, 3.625)	1 3/8 + 1 3/8 + 7/8 = 2 13/8 = 2 + 1 5/8 = 3 5/8 inches.
e2_s1	Set 2: How much taller is the 1 3/4-inch snail than a 1 1/8-inch snail? [5/8 inch / 1/2 inch / 3/4 inch / 2 7/8 inches]	**5/8 inch**	Rename 1 3/4 as 1 6/8. 1 6/8 − 1 1/8 = 5/8 inch. (2 7/8 comes from adding.)
e3_s1	Set 2: Two steps: how much more is the total height of the two 1 1/8-inch snails than the height of the tallest snail? Inches:	**1/2** (also accepted, with or without the unit word: 2/4, 4/8, 8/16, 0.5)	Two 1 1/8-inch snails: 2 2/8 = 2 1/4 inches. Tallest snail: 1 3/4 inches. Rename 2 1/4 as 1 5/4: 1 5/4 − 1 3/4 = 2/4 = 1/2 inch.
e1_s2	Set 3: *Use Mr. Varga's anemone line plot again. [Line plot "Aggregating Anemone Widths", Width (inches); X's: 1/4 (1); 1/2 (3); 5/8 (3); 3/4 (2); 1 (2); 1 1/4 (1).]* What is the combined width of the three 5/8-inch anemones and one 3/4-inch anemone? Inches:	**2 5/8** (also accepted, with or without the unit word: 2 10/16, 21/8, 2-5/8, 2 and 5/8, 2.625)	Rename 3/4 as 6/8. 5/8 + 5/8 + 5/8 + 6/8 = 21/8 = 2 5/8 inches.

ID	Question	Answer	Why (explain to student)
e2_s2	Set 3: How much wider is the 1 1/4-inch anemone than a 1/2-inch anemone? [3/4 inch / 1/2 inch / 1 3/4 inches / 7/8 inch]	**3/4 inch**	Rename 1 1/4 as 5/4 and 1/2 as 2/4. 5/4 − 2/4 = 3/4 inch. (1 3/4 comes from adding.)
e3_s2	Set 3: Two steps: how much more is the total width of the two 3/4-inch anemones than the width of the widest anemone? Inches:	**1/4** (also accepted, with or without the unit word: 2/8, 4/16, 0.25)	Two 3/4-inch anemones: 6/4 = 1 2/4 inches. Widest anemone: 1 1/4 inches. 1 2/4 − 1 1/4 = 1/4 inch.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: "A line plot shows data as X marks above a number line." What does each X on a line plot stand for? [One measurement or piece of data / One tick mark / One inch / The total of all the data]	**One measurement or piece of data**	Each X is one piece of data. Five crabs measured 1 1/2 inches means five X's stacked above 1 1/2.
v2_s0	Set 1: Type your answer: "1/2 and 4/8 are ___ fractions, so they share one stack of X's." (one word)	**equivalent** (also accepted: equal)	Fractions that name the same amount, like 1/2 and 4/8, are equivalent fractions.
v1_s1	Set 2: From the lesson: "The fair share is the amount each jar would hold if all the water were poured together and shared equally." How do you find a fair share? [Add all the data, then divide by how many pieces of data there are / Subtract the smallest value from the largest value / Find the tallest stack of X's / Count the tick marks on the number line]	**Add all the data, then divide by how many pieces of data there are**	The fair share is the total divided equally: add every value, then divide by the number of X's.
v2_s1	Set 2: Type your answer: "Each tick ___ on the number line stands for 1/8 inch." (one word)	**mark** (also accepted with "a" or "an" or in the plural)	The small marks on a number line are tick marks. On an eighths number line, each space between tick marks is 1/8.
v1_s2	Set 3: From the lesson: "The intertidal zone is the strip of shore that is underwater at high tide and uncovered at low tide." When can the team survey the tide pools in the intertidal zone? [At low tide, when the pools are uncovered / At high tide, when the pools are underwater / Only at night / Only in winter]	**At low tide, when the pools are uncovered**	The intertidal zone is uncovered at low tide, so that is when people can safely walk out and measure.
v2_s2	Set 3: Type your answer: "Before you add or subtract, rewrite the fractions with a common ___." (one word)	**denominator** (also accepted with "a" or "an" or in the plural)	To add 1/4 and 3/8, rewrite 1/4 as 2/8 so both fractions have the common denominator 8.

YOUR TURN: FAIR SHARES (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: *[Line plot "Seawater in 6 Jars", Amount (cups); X's: 1/4 (1); 1/2 (1); 5/8 (1); 7/8 (1); 1 (1); 1 1/4 (1).]* Two steps: if the water in these 6 jars were shared equally, how much would each jar hold? Cups:	**3/4** (also accepted, with or without the unit word: 6/8, 12/16, 0.75)	Total in eighths: 2/8 + 4/8 + 5/8 + 7/8 + 8/8 + 10/8 = 36/8 = 4 1/2 cups. Share among 6 jars: 36/8 ÷ 6 = 6/8 = 3/4 cup.

ID	Question	Answer	Why (explain to student)
t2_s0	Set 1: After Ms. Uchida shares some water equally, each of 5 jars holds 3/4 cup. How much water is there in all? [3 3/4 cups / 3/20 cup / 5 3/4 cups / 15/20 cup]	**3 3/4 cups**	Working backward: 5 jars × 3/4 cup = 15/4 = 3 3/4 cups. (5 3/4 comes from adding 5 + 3/4.)
t1_s1	Set 2: *[Line plot "Seawater in 8 Jars", Amount (cups); X's: 1/4 (1); 3/8 (1); 1/2 (1); 5/8 (1); 3/4 (2); 7/8 (2).]* Two steps: if the water in these 8 jars were shared equally, how much would each jar hold? Cups:	**5/8** (also accepted: 5/8 cups, 5/8 cup, 5/8 c, 10/16, 10/16 cups, 10/16 cup, 10/16 c, 0.625, 0.625 cups, 0.625 cup, 0.625 c)	Total in eighths: 2/8 + 3/8 + 4/8 + 5/8 + 6/8 + 6/8 + 7/8 + 7/8 = 40/8 = 5 cups. Share among 8 jars: 5 ÷ 8 = 5/8 cup.
t2_s1	Set 2: After Ms. Uchida shares some water equally, each of 4 jars holds 7/8 cup. How much water is there in all? [3 1/2 cups / 7/32 cup / 4 7/8 cups / 28/32 cup]	**3 1/2 cups**	Working backward: 4 jars × 7/8 cup = 28/8 = 3 4/8 = 3 1/2 cups. (4 7/8 comes from adding 4 + 7/8.)
t1_s2	Set 3: *[Line plot "Seawater in 5 Jars", Amount (cups); X's: 0 1/8 (1); 3/8 (1); 1/2 (1); 5/8 (1); 7/8 (1).]* Two steps: if the water in these 5 jars were shared equally, how much would each jar hold? Cups:	**1/2** (also accepted, with or without the unit word: 2/4, 4/8, 8/16, 0.5)	Total in eighths: 1/8 + 3/8 + 4/8 + 5/8 + 7/8 = 20/8 = 2 1/2 cups. Share among 5 jars: 20/8 ÷ 5 = 4/8 = 1/2 cup.
t2_s2	Set 3: After Ms. Uchida shares some water equally, each of 6 jars holds 3/8 cup. How much water is there in all? [2 1/4 cups / 3/48 cup / 6 3/8 cups / 18/48 cup]	**2 1/4 cups**	Working backward: 6 jars × 3/8 cup = 18/8 = 2 2/8 = 2 1/4 cups. (6 3/8 comes from adding 6 + 3/8.)

REAL WORLD: VOLUNTEER SHIFTS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation On busy low-tide mornings, volunteers stand near the tide pools to answer questions and remind visitors not to climb the rock. Mr. Varga made a line plot of how long 10 volunteers worked on Saturday. [Line plot "Volunteer Shift Lengths (Saturday)", Time (hours); X's: 1/2 (2); 3/4 (3); 1 (2); 1 1/4 (2); 1 1/2 (1).]* Two steps: how many hours did the 3/4-hour volunteers and the 1 1/4-hour volunteers work altogether? Hours:	**4 3/4** (also accepted, with or without the unit word: 4 6/8, 4 12/16, 19/4, 4-3/4, 4 and 3/4, 4.75)	Three volunteers × 3/4 = 9/4 hours. Two volunteers × 1 1/4 = 2 2/4 = 10/4 hours. 9/4 + 10/4 = 19/4 = 4 3/4 hours.
w1_s1	Set 2: *The Situation On busy low-tide mornings, volunteers stand near the tide pools to answer questions and remind visitors not to climb the rock. Mr. Varga made a line plot of how long 10 volunteers worked on Sunday. [Line plot "Volunteer Shift Lengths (Sunday)", Time (hours); X's: 1/4 (1); 1/2 (3); 3/4 (2); 1 (3); 1 1/2 (1).]* Two steps: how many more hours did the 1-hour volunteers work in all than the 1/2-hour volunteers worked in all? [1 1/2 hours / 1/2 hour / 4 1/2 hours / 1 hour]	**1 1/2 hours**	Three volunteers × 1 hour = 3 hours. Three volunteers × 1/2 hour = 1 1/2 hours. 3 − 1 1/2 = 1 1/2 hours. (1/2 hour compares just one volunteer from each group, and 4 1/2 comes from adding.)

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: *The Situation On busy low-tide mornings, volunteers stand near the tide pools to answer questions and remind visitors not to climb the rock. Mr. Varga made a line plot of how long 8 volunteers worked on Monday. [Line plot "Volunteer Shift Lengths (Monday)", Time (hours); X's: 1/2 (2); 3/4 (2); 1 (1); 1 1/4 (2); 2 (1).]* Two steps: if the total time were shared equally among all 8 volunteers, how many hours would each one work? Hours:	**1** (also accepted: 1 hours, 1 hour, 1 hr)	Total in fourths: 2/4 + 2/4 + 3/4 + 3/4 + 4/4 + 5/4 + 5/4 + 8/4 = 32/4 = 8 hours. Share among 8 volunteers: 8 ÷ 8 = 1 hour each.

REAL WORLD: COLLECTING AND CLEANING UP (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Mr. Varga needs 4 cups of seawater for a lab test back at the school. So far the team has filled 6 jars. [Line plot "Seawater Collected So Far", Amount (cups); X's: 1/4 (1); 3/8 (2); 1/2 (1); 5/8 (1); 3/4 (1).]* Three steps: how much more seawater does the team need to collect? [1 1/8 cups / 2 7/8 cups / 1 7/8 cups / 6 7/8 cups]	**1 1/8 cups**	Total in eighths: 2/8 + 3/8 + 3/8 + 4/8 + 5/8 + 6/8 = 23/8 = 2 7/8 cups. Rename 4 as 3 8/8: 3 8/8 − 2 7/8 = 1 1/8 cups. (2 7/8 is what they already have, not what they still need.)
w2_s1	Set 2: *The Situation Taking litter off the beach is allowed and helpful. The cleanup crew's goal is 8 pounds of litter. Uzoma weighed each crew member's bag. [Line plot "Litter Collected per Bag", Weight (pounds); X's: 1/2 (2); 3/4 (3); 1 (1); 1 1/4 (2).]* Three steps: how many more pounds of litter does the crew need to reach the goal? Pounds:	**1 1/4** (also accepted, with or without the unit word: 1 2/8, 1 4/16, 5/4, 1-1/4, 1 and 1/4, 1.25)	Total in fourths: 2/4 + 2/4 + 3/4 + 3/4 + 3/4 + 4/4 + 5/4 + 5/4 = 27/4 = 6 3/4 pounds. Rename 8 as 7 4/4: 7 4/4 − 6 3/4 = 1 1/4 pounds.
w2_s2	Set 3: *The Situation Each morning, the team walks a line along the beach and counts sea stars. Mr. Varga plotted the length of 9 walks. [Line plot "Sea Star Count Walks", Distance (miles); X's: 1/4 (2); 3/8 (3); 1/2 (2); 5/8 (1); 3/4 (1).]* Two steps: what is the total distance of all the walks that were shorter than 1/2 mile? Miles:	**1 5/8** (also accepted, with or without the unit word: 1 10/16, 13/8, 1-5/8, 1 and 5/8, 1.625)	Shorter than 1/2: two walks at 1/4 and three at 3/8. 2 × 1/4 = 2/4 = 4/8. 3 × 3/8 = 9/8. 4/8 + 9/8 = 13/8 = 1 5/8 miles.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? Six sample jars hold a total of 3 cups of seawater. What is the fair share for each jar? [1/2 cup, because 3 ÷ 6 = 3/6 = 1/2 / 2 cups, because 6 ÷ 3 = 2 / 3 cups, because the total stays the same / 1/6 cup, because there are 6 jars]	**1/2 cup, because 3 ÷ 6 = 3/6 = 1/2**	The fair share is the total divided by the number of jars: 3 ÷ 6 = 3/6 = 1/2 cup. Dividing 6 ÷ 3 puts the numbers in the wrong order.

ID	Question	Answer	Why (explain to student)
x2_s0	Set 1: Find the mistake: Uzoma adds two crab widths and writes 3/8 + 1/4 = 4/12. What should he fix? [He added the denominators. Rename 1/4 as 2/8: 3/8 + 2/8 = 5/8 / He should multiply: 3/8 × 1/4 = 3/32 / He should subtract: 3/8 − 1/4 = 1/8 / Nothing. 4/12 is correct]	**He added the denominators. Rename 1/4 as 2/8: 3/8 + 2/8 = 5/8**	When you add fractions, the denominator names the size of the pieces and stays the same. Rewrite with a common denominator first: 3/8 + 2/8 = 5/8.
x1_s1	Set 2: Which explanation is correct? Where does the X for 6/8 inch go on a line plot marked in eighths and labeled with simplest fractions? [On the tick labeled 3/4, because 6/8 and 3/4 are equivalent / Above the number 6, because the numerator is 6 / Between 3/4 and 7/8, because 6/8 is bigger than 3/4 / Nowhere, because 6/8 is not in simplest form]	**On the tick labeled 3/4, because 6/8 and 3/4 are equivalent**	6/8 = 3/4 (divide the top and bottom by 2), so it is the same point on the number line. The X goes in the 3/4 stack.
x2_s1	Set 2: Find the mistake: Vikram says a 2-inch limpet is 3/8 inch longer than a 5/8-inch limpet. What should he fix? [Rename 2 as 1 8/8: 1 8/8 − 5/8 = 1 3/8 inches / He should write 2 − 5/8 = 1 5/8 inches / Nothing. 3/8 inch is correct / He should add: 2 + 5/8 = 2 5/8 inches]	**Rename 2 as 1 8/8: 1 8/8 − 5/8 = 1 3/8 inches**	To subtract 5/8 from 2, regroup one whole as 8/8: 2 = 1 8/8, and 1 8/8 − 5/8 = 1 3/8. Vikram found 8/8 − 5/8 = 3/8 but forgot the 1 whole that was left.
x1_s2	Set 3: Which explanation is correct? A line plot shows 4 X's above 3/4 inch. What is the total length of those 4 shells? [3 inches, because 4 × 3/4 = 12/4 = 3 / 3/16 inch, because 3/4 ÷ 4 = 3/16 / 4 3/4 inches, because 4 + 3/4 = 4 3/4 / 12/16 inch, because 4 × 3 = 12 and 4 × 4 = 16]	**3 inches, because 4 × 3/4 = 12/4 = 3**	Each X is one shell that is 3/4 inch long, so add 3/4 four times: 4 × 3/4 = 12/4 = 3 inches. Only the numerator is multiplied.
x2_s2	Set 3: Find the mistake: five jars hold 1/4, 1/2, 1/2, 3/4, and 1 cup. Uzoma adds them to get 3 cups, then divides by 4 because the line plot has 4 stacks. He says the fair share is 3/4 cup. What should he fix? [Divide by the 5 jars, not the 4 stacks: 3 ÷ 5 = 3/5 cup / Nothing. 3/4 cup is correct / Divide by 3 to get 1 cup / Multiply 3 × 5 to get 15 cups]	**Divide by the 5 jars, not the 4 stacks: 3 ÷ 5 = 3/5 cup**	Each X is one jar. There are 5 jars, so the fair share is 3 ÷ 5 = 3/5 cup. The number of stacks does not matter.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: vikram asked, which tide pool should we measure first	**Vikram asked, "Which tide pool should we measure first?"**	Capitalize the name Vikram, put quotation marks around his exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: Haystack Rock is 235 feet tall, and it's base is covered with mussels.	**Haystack Rock is 235 feet tall, and its base is covered with mussels.**	Its (no apostrophe) shows ownership: the base belongs to the rock. It's always means it is.
s3_s0	Set 1: Fix this sentence: At low tide the team measured limpets snails and crabs.	**At low tide, the team measured limpets, snails, and crabs.** (also accepted: At low tide, the team measured limpets, snails and crabs.)	Put a comma after the opening phrase At low tide. Then use commas to separate the items in a series: limpets, snails, and crabs.

ID	Question	Answer	Why (explain to student)
s4_s0	Set 1: Fix this sentence: the sea stars on the north side of the rock was easy to count	**The sea stars on the north side of the rock were easy to count.**	Capitalize the first word. The subject is sea stars (plural), not rock, so use were, not was. End with a period.
s1_s1	Set 2: Fix this sentence: ms. uchida said, never take a shell from the tide pools	**Ms. Uchida said, "Never take a shell from the tide pools."**	Capitalize Ms. and the name Uchida, put quotation marks around her exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: Their measuring the anemones before the tide comes in.	**They're measuring the anemones before the tide comes in.**	They're is short for they are: they are measuring. Their shows ownership.
s3_s1	Set 2: Fix this sentence: We visited Cannon Beach Oregon on June 14 2026.	**We visited Cannon Beach, Oregon, on June 14, 2026.**	Put a comma between the city and the state, and another after the state. Put a comma between the day and the year.
s4_s1	Set 2: Fix this sentence: each of the jars hold less than one cup of seawater	**Each of the jars holds less than one cup of seawater.**	Capitalize the first word. The subject is each (one jar at a time), so use holds, not hold. End with a period.
s1_s2	Set 3: Fix this sentence: uzoma shouted, look at the puffins on the rock	**Uzoma shouted, "Look at the puffins on the rock!"** (also accepted: Uzoma shouted, "Look at the puffins on the rock.")	Capitalize the name Uzoma, put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark (an exclamation point fits a shout) inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Mr. Varga said that your going to need a ruler.	**Mr. Varga said that you're going to need a ruler.**	You're is short for you are: you are going to need a ruler. Your shows ownership, as in your ruler.
s3_s2	Set 3: Fix this sentence: Before the tide came in Vikram recorded his last measurement.	**Before the tide came in, Vikram recorded his last measurement.**	Put a comma after the opening part, Before the tide came in, to show where the main sentence starts.
s4_s2	Set 3: Fix this sentence: the line plot show three crabs at 1 3/8 inches	**The line plot shows three crabs at 1 3/8 inches.**	Capitalize the first word. The subject is the line plot (one plot), so use shows, not show. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 3/8 + 1/4	**5/8**	2 points if matched on the first try.
match_s0_2	Set 1: Match 2 − 3/4	**1 1/4**	2 points if matched on the first try.
match_s0_3	Set 1: Match 4 × 3/8	**1 1/2**	2 points if matched on the first try.
match_s0_4	Set 1: Match 3 cups shared by 4 jars	**3/4 cup**	2 points if matched on the first try.
match_s0_5	Set 1: Match 1/2 + 3/8	**7/8**	2 points if matched on the first try.
match_s1_1	Set 2: Match 1/4 + 5/8	**7/8**	2 points if matched on the first try.
match_s1_2	Set 2: Match 1 1/2 − 7/8	**5/8**	2 points if matched on the first try.
match_s1_3	Set 2: Match 3 × 3/4	**2 1/4**	2 points if matched on the first try.
match_s1_4	Set 2: Match 2 cups shared by 8 jars	**1/4 cup**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s1_5	Set 2: Match 5/8 + 3/4	**1 3/8**	2 points if matched on the first try.
match_s2_1	Set 3: Match 1/2 + 1/4	**3/4**	2 points if matched on the first try.
match_s2_2	Set 3: Match 2 − 1 1/8	**7/8**	2 points if matched on the first try.
match_s2_3	Set 3: Match 5 × 1/4	**1 1/4**	2 points if matched on the first try.
match_s2_4	Set 3: Match 5 cups shared by 8 jars	**5/8 cup**	2 points if matched on the first try.
match_s2_5	Set 3: Match 3/8 + 3/4	**1 1/8**	2 points if matched on the first try.

BOSS BATTLE: THE SURVEY REPORT (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Vikram and Mr. Varga Write the Report At the end of the week, Vikram plots the shell widths of 10 purple shore crabs from a new tide pool. [Line plot "Crab Shell Widths, New Tide Pool", Width (inches); X's: 1 1/4 (1); 1 3/8 (2); 1 1/2 (3); 1 5/8 (2); 1 3/4 (1); 2 (1).]* Two steps: how much more is the total width of the three 1 1/2-inch crabs than the total width of the two 1 5/8-inch crabs? [1 1/4 inches / 1/8 inch / 7 3/4 inches / 1 3/4 inches]	**1 1/4 inches**	Three crabs × 1 1/2 = 4 1/2 inches. Two crabs × 1 5/8 = 2 10/8 = 3 2/8 = 3 1/4 inches. 4 1/2 − 3 1/4 = 4 2/4 − 3 1/4 = 1 1/4 inches. (1/8 inch compares only one crab from each stack.)
b2_s0	Set 1: What is the total width of all 10 crabs? Inches:	**15 1/2** (also accepted, with or without the unit word: 15 2/4, 15 4/8, 15 8/16, 31/2, 15-1/2, 15 and 1/2, 15.5)	In eighths: 1 1/4 = 10/8, 1 3/8 = 11/8, 1 1/2 = 12/8, 1 5/8 = 13/8, 1 3/4 = 14/8, 2 = 16/8. Total: 10 + 2(11) + 3(12) + 2(13) + 14 + 16 = 124 eighths. 124/8 = 15 4/8 = 15 1/2 inches.
b1_s1	Set 2: *Vikram and Ms. Uchida Write the Report On the last morning, Ms. Uchida fills 8 jars with seawater for testing. Vikram plots how full each jar is. [Line plot "Seawater in 8 Jars, Last Morning", Amount (cups); X's: 1/4 (1); 3/8 (2); 1/2 (2); 3/4 (2); 1 1/2 (1).]* Two steps: if all the water were shared equally among the 8 jars, how much would each jar hold? [5/8 cup / 1 cup / 5 cups / 1 3/5 cups]	**5/8 cup**	Total in eighths: 2 + 3 + 3 + 4 + 4 + 6 + 6 + 12 = 40 eighths = 5 cups. Share among 8 jars: 5 ÷ 8 = 5/8 cup. (1 cup comes from dividing by the 5 stacks instead of the 8 jars.)
b2_s1	Set 2: Two steps: Mr. Varga pours the 1 1/2-cup jar and both 3/4-cup jars into an empty 4-cup bottle. How many more cups of water will fit in the bottle? Cups:	**1** (also accepted: 1 cups, 1 cup, 1 c)	Poured in: 1 1/2 + 3/4 + 3/4 = 1 1/2 + 1 1/2 = 3 cups. Room left: 4 − 3 = 1 cup.
b1_s2	Set 3: *Vikram and Uzoma Write the Report Vikram and Uzoma plot the lengths of all 9 sea star count walks from their second week. [Line plot "Sea Star Count Walks, Second Week", Distance (miles); X's: 1/4 (2); 3/8 (1); 1/2 (3); 3/4 (2); 1 (1).]* Two steps: how much farther did the three 1/2-mile walks cover in all than the two 1/4-mile walks and the 3/8-mile walk together? [5/8 mile / 3/8 mile / 2 3/8 miles / 1/4 mile]	**5/8 mile**	Three 1/2-mile walks: 1 1/2 = 12/8 miles. Two 1/4-mile walks and one 3/8-mile walk: 4/8 + 3/8 = 7/8 mile. 12/8 − 7/8 = 5/8 mile. (2 3/8 comes from adding the two groups.)

ID	Question	Answer	Why (explain to student)
b2_s2	Set 3: What is the total distance of all 9 walks? Miles:	**4 7/8** (also accepted, with or without the unit word: 4 14/16, 39/8, 4-7/8, 4 and 7/8, 4.875)	In eighths: 2 × 2/8 + 3/8 + 3 × 4/8 + 2 × 6/8 + 8/8 = 4 + 3 + 12 + 12 + 8 = 39 eighths. 39/8 = 4 7/8 miles.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W23.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: a) 2 1/3 yd to feet b) 2 gal 3 qt to quarts c) 3.2 kg to grams	a) 6 + 1 = 7 feet b) 8 + 3 = 11 quarts c) 3.2 × 1,000 = 3,200 grams
2	Circle every fraction equal to 1/2	2/4, 4/8 and 3/6 (not 3/8 or 5/8).
3	Add, renaming first: a) 3/8 + 1/4 b) 1/2 + 3/8	a) 3/8 + 2/8 = 5/8 b) 4/8 + 3/8 = 7/8
4	Number line 0 to 1 in eighths; label 1/4, 1/2, 3/4	Eight equal spaces. 1/4 is on the 2nd tick (2/8), 1/2 on the 4th (4/8), and 3/4 on the 6th (6/8).
5	Six limpets: 1/2, 3/4, 1/2, 5/8, 3/4, 1/2 inch	1/2 inch shows up most often, 3 times.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Line plot in eighths of 8 anemone widths	Rename 4/8 = 1/2 and 6/8 = 3/4. X's: 3/8 → 1, 1/2 → 3, 5/8 → 1, 3/4 → 2, 7/8 → 1 (8 X's in all). Check for a title and the label "inches."
2	a) How many wider than 1/2 inch? b) Widest minus narrowest	a) 4 anemones (1 at 5/8, 2 at 3/4, 1 at 7/8). b) 7/8 − 3/8 = 4/8 = 1/2 inch.
3	Six jars: 1/4, 3/8, 1/2, 3/4, 7/8 and 1 cup; total and fair share	Total in eighths: 2/8 + 3/8 + 4/8 + 6/8 + 7/8 + 8/8 = 30/8 = 3 6/8 = 3 3/4 cups. Fair share: 30/8 ÷ 6 = 5/8 cup.
4	Uzoma says 6 cups shared by 8 jars is 1 1/3 cups each	Model: Uzoma divided in the wrong order, 8 ÷ 6 instead of 6 ÷ 8. The fair share is the total divided by the number of jars, so 6 ÷ 8 = 6/8 = 3/4 cup. A share of more than 1 cup would need more than 8 cups of water.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	4 X's at 5/8, 3 X's at 3/4, 1 X at 1 1/4: total and fair share	In eighths: 4 × 5/8 = 20/8, 3 × 3/4 = 18/8, 1 1/4 = 10/8. Total: 48/8 = 6 inches. Fair share: 6 ÷ 8 = 6/8 = 3/4 inch.